Slayer1B: Continuing a Polish-Dominant Base Model from 1.5 to 3 Billion Tokens

SlayerLab

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Evidence cutoff: 23 October 2026, 21:41 UTC. Completed releases 1.00 and 1.01; the 6-billion-token continuation is an ongoing experiment, not a reported result. This is a complete preliminary technical report, not a claim of architectural superiority.

Abstract

We document a Polish-dominant decoder-only language model and a continuation experiment that preserves its architecture and resumes the full optimizer state before learning-rate decay. The longer-trained release improves English and Polish science-question answering, English grammatical acceptability, and English language-model loss under one fixed evaluation instrument. The Polish grammatical-acceptability difference remains unresolved by the paired interval. A separately labelled evaluation protocol adds commonsense and reading diagnostics; several of its differences also remain unresolved. We distinguish model-token history from newly executed optimization, measured scores from conditional loss scenarios, and verified artifact bytes from repository metadata. The evidence supports the usefulness of this continuation recipe on these tests. It does not isolate an architectural effect, establish broad usefulness, or support a state-of-the-art claim.

1 Question and scope

The operational question is whether a short-budget base model benefits from extending training while postponing its final learning-rate decay. Release 1.01 continues release 1.00 from a saved pre-decay state, rather than reheating its cooled final state. The experiment changes the training horizon and eventually supplies additional data, but does not change model size or architecture.

"1B" is a model-family name: the actual count is **1,073,912,851 trainable parameters**, with tied embeddings counted once. This exceeds a strict one-billion-parameter eligibility limit. All results here concern raw base models: no instruction tuning, safety tuning, or task-specific adaptation was applied to these releases.

2 Model and optimization

The model uses established Transformer components. This report does not introduce them as novel contributions or identify their individual effects at this scale. The tokenizer is unchanged between releases; end-of-sequence token ID is 0. The inference package is a native PyTorch GPT implementation with safetensors weights, not a Transformers `AutoModel` checkpoint.

2.1 Schedule and indexing

Let s be the zero-based update index, T the intended total number of completed updates, $w = 150$, and $q = \lfloor 0.8T \rfloor$. For the AdamW learning rate, the implemented schedule is

$$\eta(s) = \begin{cases} \eta_{\max}(s+1)/w, & s < w, \\ \eta_{\max}, & w \le s < q, \\ \eta_{\min} + (\eta_{\max} - \eta_{\min})\left(1 - \sqrt{(s-q)/(T-q)}\right), & q \le s < T, \end{cases}$$

where $\eta_{\max} = 6 \times 10^{-4}$ and $\eta_{\min} = 6 \times 10^{-5}$. At $s = q$ the decay fraction is zero, so that update still uses the peak rate. Logs and checkpoint names count completed updates, $s+1$; a decay-phase

Property	Configuration
Transformer	Decoder-only; 20 blocks; width 2048
Attention	16 full attention heads; head dimension 128
Position and normalization	RoPE, $\theta = 500000$; pre-norm RMSNorm, $\epsilon = 10^{-6}$
Additional components	QK normalization; value residual
Feed-forward	SwiGLU; hidden width 5462; no linear biases
Vocabulary / context	32,768 byte-level BPE tokens / 4096 tokens
Embeddings	Input/output weights tied
Hidden matrix optimizer	Muon; nominal peak learning rate 0.02
Other parameters	AdamW; peak learning rate 6×10^{-4}; weight decay 0.1
Training execution	8 RTX 5090 GPUs, DDP; bf16 autocast
Memory / compilation	Activation checkpointing; `torch.compile`
Global batch	$256 \times 4096 = 1,048,576$ tokens per update

Table 1: Shared architecture and training configuration. The released configuration and trainer are the executable specification, not an implementation inferred from the model name.

boundary must not be mistaken for the first lower-rate update. For release 1.01, $q = 2288$, peak rate continues through completed update 2289, and the first lower-rate update is 2290. The final update uses approximately 6.05×10^{-5}, since the loop ends at $s = T - 1$.

2.2 Resume and compute accounting

Version	Resume	Final T	Model-history tokens	Newly executed updates
1.00, completed	0	1430	1,499,463,680	1430
1.01, completed	1050	2860	2,998,927,360	1810
1.02, planned	2250	5720	5,997,854,720	3470

Table 2: The final column is the new work after each resume, not the difference between released final token counts. The 1.02 row is a plan, not an achieved budget.

Release 1.01 starts from the last saved checkpoint before the 1.00 decay. Its newly executed work is 1,897,922,560 tokens, because the cooled 1.00 tail is not inherited. The common prefix is retained; the discarded tail still consumed project compute. Consequently, model-history tokens are not a complete project-compute ledger. The two completed runs together executed 3,397,386,240 training tokens, excluding setup and synthetic checks.

Native checkpoints contain model weights, optimizer state, completed step, and configuration. Data order is reconstructed from the step and fixed data seed 20261023. We do not claim that these files explicitly serialize all Python, NumPy, and CUDA random-number states. Two checks of early continuation windows report signed mean loss differences of -0.00008 nats per token over updates 1060–1140 and -0.00013 over updates 1060–1130; the reported maximum absolute single-point difference is 0.0007. These are observed numerical-continuity checks, not a promise of bitwise identity on arbitrary hardware.

The archived 1.01 launch arguments record eight gradient-accumulation steps and a 1024-token loss chunk. The public checkpoint repository includes these arguments, software freeze, training log, and data manifests; the resumed native step, rather than an argument-file label alone, establishes the actual continuation point.

3 Data and token units

The source dataset is `SlayerLab/slayer-pl-8x3b` at revision `d50df04a6ad46876fa559a87cca5d3e2a96e7098`. The pack mixing weights are defined in `cl100k` token units. They are not fractions of this model's tokenizer output.

Component	Sampling weight	Pack-00 native tokens	Pack-01 native tokens
Polish web	63%	51.79%	52.16%
Register-diverse core	19%	22.19%	22.14%
Legal / public administration	8%	8.46%	8.20%
English	10%	17.56%	17.49%

Table 3: Native shares refer to the training split, not training plus validation. Rounded shares can differ slightly from 100%. Tokenization makes a nominal 10% English mixing weight a substantially larger share of native training tokens.

Pack-00 supplies 1,889,519,627 available training tokens; pack-01 supplies 1,880,520,445. The epoch sampler uses each selected 4096-token window index at most once within the relevant stream. This is a statement about sampling indices, not proof that the underlying text has no duplicate or semantically similar passages.

Release 1.01 follows pack-00 through completed update 1801, then pack-01 from update 1802 through 2860. The trainer's `-switch-step 1801` is zero-based; labelling completed update 1801 as the first new-pack batch would be an off-by-one error. The pack-01 segment consumes 1059 global batches. Web slices are disjoint by construction; the shared core, legal, and English components use non-overlapping document-hash intervals between packs.

The same rule-based masking is applied before tokenization: email, telephone, PESEL, NIP, REGON, and Polish identity-card patterns become typed tags. This does not establish removal of every personal datum. Dataset components retain their own licenses, including CC-BY-SA partitions; the Apache-2.0 weights license must not be read as relicensing the data.

A reported 13-gram overlap diagnostic for pack-01 found no evaluation item with at least 90% coverage. Ten of 3272 Polish MultiBLiMP items had at least 50% coverage. Twenty-six of 62 WikiText-2 documents shared at least one 13-gram, but none reached 50% coverage. These checks do not prove contamination immunity or exclude semantic overlap.

4 Primary evaluation: one fixed instrument

The primary instrument is `glint_parity_eval.py`; its full archived fingerprint is listed with the release artifacts below. All checkpoints use its same scoring conventions and fp32 inference. English BLiMP uses summed log-probabilities; ARC-Easy reports raw accuracy. WikiText-2 uses non-overlapping 256-token chunks on a fixed raw test text of 1,292,008 UTF-8 bytes. Polish MultiBLiMP uses the fixed leading-space convention and summed log-probabilities; ARC-Easy-PL uses $\log P(q + \text{choice}) - \log P(q)$.

The local leaderboard also derives an `eff` index from English BLiMP, ARC-Easy, and WikiText-2 byte perplexity. It is a summary of those measurements, not an additional benchmark or a general usefulness score. We do not use it to replace the individual task results. A common held-out Polish language-model score is not reported: its required corpus was unavailable at this evidence cutoff; missing is not zero.

Task	Metric	Items	1.00	1.01
ARC-Easy, English	Accuracy (%)	2376	42.89	47.98
BLiMP, English	Accuracy (%)	67000	71.51	73.63
WikiText-2	Byte perplexity	fixed text	2.5857	2.4230
WikiText-2	Native-token perplexity	same text	49.9734	38.2380
MultiBLiMP, Polish	Accuracy (%)	3272	97.89	98.29
ARC-Easy-PL	Accuracy (%)	2376	36.91	40.36

Table 4: Final released checkpoints, 1430 and 2860. Byte and token perplexity are two normalizations of one language-model measurement, not independent capabilities. English and translated Polish ARC are related tasks.

4.1 Paired differences and uncertainty

Retained per-item outcomes permit paired bootstrap intervals: 10,000 resamples, seed 0, using the same item identities for both models. Intervals are pointwise and unadjusted for multiple comparisons. They describe these evaluation outcomes, not variance across training seeds or execution environments.

Task	Difference (pp)	95% interval (pp)	Interpretation
ARC-Easy, English	+5.093	[3.451, 6.776]	Positive on this test
BLiMP, English	+2.124	[1.831, 2.427]	Positive on this test
ARC-Easy-PL	+3.451	[1.852, 5.008]	Positive on this test
MultiBLiMP, Polish	+0.397	[−0.061, 0.856]	Unresolved

Table 5: Differences are computed from unrounded outcomes. No interval is claimed for the single fixed WikiText-2 text. A near-ceiling Polish grammar score is not evidence that broad Polish capability is solved.

The baseline remeasurement counted 47,912 correct English BLiMP items; an older execution on a different GPU counted 47,913, with the same displayed percentage. The old execution did not preserve per-item decisions, so the cause and number of changed item decisions are unknown. Two independent implementations and a repeated execution of the retained baseline agree item by item. The paired comparison uses the retained outcomes, not an invented reconstruction of the older run. Baseline and final measurements were not all made on one physical host.

5 Supplementary evaluation: a separate protocol

A native-model adapter also evaluates the releases with `lm-eval` 0.4.13, batch size one and native context ceilings. The completed 1.01 measurement uses PyTorch 2.8.0+cu128, Transformers 5.20.2, Hugging Face Hub 1.33.0, tokenizers 0.23.2, datasets 5.1.0, and accelerate 1.15.0. This protocol differs in prompting, task implementations, and language-model scoring from the primary instrument. Its absolute scores must not be spliced into the preceding table or leaderboard.

For completeness, supplementary raw ARC-Easy accuracy is 41.79% versus 46.34%, and raw ARC-Challenge accuracy is 17.49% versus 20.05%; those are different metrics from `acc_norm`. The baseline English execution and final execution differ in host and software environment. Polish baseline calibration had reproduced the earlier outcomes; a repeat under the final environment was still in progress at cutoff. These limitations prevent treating the two protocols as independent replications of a tightly controlled training intervention. Additional external-model comparisons were still pending and are not represented by substituted vendor scores.

Task	Metric	1.00	1.01	Paired 95% interval, pp
ARC-Easy	`acc_norm`, %	38.59	41.29	$[1.18, 4.17]$
ARC-Challenge	`acc_norm`, %	21.42	24.32	$[1.11, 4.69]$
HellaSwag	`acc_norm`, %	28.59	30.80	$[1.62, 2.80]$
PIQA	`acc_norm`, %	57.62	58.16	$[-1.03, 2.18]$
WinoGrande	`acc`, %	50.12	51.70	$[-1.66, 4.81]$
BLiMP	`acc`, %	73.82	76.16	$[2.04, 2.64]$
Belebele, Polish	`acc`, %	23.22	26.00	$[-0.11, 5.78]$
MultiBLiMP, Polish	`acc`, %	97.95	98.41	$[-0.03, 0.95]$
WikiText-2	Byte perplexity	2.0706	1.9591	not estimated
WikiText-2	Word perplexity	49.01	36.46	not estimated

Table 6: Supplementary measurements, not the primary leaderboard protocol. Intervals concern the paired difference, with the same bootstrap count and seed as above. Four accuracy intervals include zero. Belebele has four choices and a 25% chance reference; this result does not establish reliable reading comprehension.

6 What the evidence explains—and does not

Thesis. The released recipe can continue learning usefully at this short token budget: several held-out diagnostics improve, not just training loss or a composite index. Fixed architecture, tokenizer, and a shared pre-decay prefix make the continuation interpretable as one model family rather than a comparison between unrelated models.

Alternative explanations and limits. Additional optimization, fresh data, and repositioned learning-rate decay change together. The experiment therefore does not isolate a pure token-count effect, an individual architecture component, or the superiority of Muon over another optimizer. No matched-parameter, matched-data, multi-seed architecture control is supplied. Public benchmark overlap and repeated inspection remain possible sources of optimistic interpretation. Strong grammar discrimination and lower next-token loss do not imply factual reliability, instruction following, mathematical reasoning, or safe dialogue.

Synthesis. The justified claim is narrower than "the architecture is better": this particular continuation recipe improves several measured capabilities of this base model. The likely explanation combines usable remaining model capacity, continued gradient updates on new windows, and a schedule that retains a stable-rate phase before its final decay. These are mechanism hypotheses consistent with the observations, not separately identified causal effects. The practical decision is to test a longer continuation while preserving the same accounting and evaluation boundaries.

7 Training-loss scenarios are not validated scaling laws

The published conditional scenarios fit the learning-rate-annealing model of Tissue et al. [7] to the 1.00 training-loss curve for several choices of its memory parameter. An earlier backtest missed measured loss by 0.077–0.088 nats per token against a 0.03 threshold. The displayed ± 0.09 tolerance is a historical-error band, not a calibrated confidence or prediction interval. The analysis reports agreement between two implementations to 3.4×10^{-7} nats; that check concerns computational consistency, not forecast validity.

For the completed continuation, the stable-rate window at updates 1700–1800 averages 2.6977 nats per token and lies near the lower edge of its prewritten band $[2.692, 2.882]$. The last ten logged values, 2770–2860, average 2.5083 within the prewritten band $[2.367, 2.660]$; including the preceding boundary point gives 2.5072. The exact end-window boundary was clarified after

seven of those ten observations were already available. Consequently, this is not a clean, fully prospective validation of the window definition. Neither the band hit nor a post-hoc closest parameter choice identifies a universal scaling law. These are training-loss diagnostics; no benchmark scores are extrapolated.

8 Artifact provenance and operational accounting

The public release separates resumable optimizer checkpoints from normalized inference weights. All 13 new native checkpoints of 1.01 were independently downloaded and hashed against the off-node source records. The normalized safetensors export was independently reproduced from the final native state with identical bytes. Repository LFS object IDs are declarations; they were not used as substitutes for hashing downloaded bodies.

Artifact	SHA-256
1.00 final native, step 1430	`7d092d4d0bf838da1115a5369bf0a6729fe4c170e0d5ad303873f7320d6c1866`
1.01 final native, step 2860	`b0a288798ae614c38e37ff8176a293f6650cec2221d59e46d5116afac7c31eba`
1.01 inference weights	`607d1a6c476bebb7f2cffb822c874b73855d26a9b8395658389f9aeff7485edd`
Tokenizer	`0640d3bd3674d7a6e59c540d945a2ad275a12e1bcaeb6007c37a90751f9815dd`
Released R23 trainer	`db36c150e95c184e82b5f84c3ee783fd5fd2c62e960a438177f543dc3eba1d0b`
Primary evaluation instrument	`d7a38ca0b7f7d2727d412912fb072e0ed0e0dd640519e48ddc8b058e777273ec`
1.02 resume state, step 2250	`aeddfb023abb11c8654b47c728c9680c232d1b6568697e3a1eee8128efd37054`

Table 7: Full artifact fingerprints. Public release and checkpoint repositories are listed in the references.

The final native 1.01 optimizer state occupies 8,860,287,151 bytes. The normalized 1.01 inference file occupies 4,295,670,676 bytes. Process-window throughput and nominal allocation cost have different denominators:

Version	Launcher to training-complete	Window-average tokens/s	Nominal main cost, USD
1.00	4 h 12 min 50 s	approximately 98,800	approximately 31.30
1.01	5 h 00 min 34 s	approximately 105,200	approximately 40.73

Table 8: The process window includes checkpointing but excludes provider provisioning and post-training offloading. The cost column instead covers the recorded full main-allocation lifetime, including setup and stopped storage; it excludes network and separate evaluation instances. It is not a cost of the process window alone.

The historical active rates, including allocated disk, were USD 6.1963/h for 1.00 and USD 7.41556/h for 1.01; stopped-storage rates were USD 0.0417/h and USD 0.0556/h. Active disk cost is not added twice. These are time-and-rate reconstructions, not provider invoices. The 1.00 deletion endpoint was human-reported; the initial active start of 1.01 was approximate. The short 1.01 restart for offloading is counted as active, not stopped, time. Account-balance changes are not attributed to one experiment, and unconfirmed bandwidth units are not interpreted as zero network cost. We make no complete project-total cost or cost-efficiency superiority

claim. The approximately 108,000 tokens/s steady-step summary is distinct from the measured process-window average.

The measured history is public at 150-update intervals plus final checkpoints: ten points for 1.00 and thirteen new points for 1.01. Training and evaluation completion are separate events; a run ending does not imply that all later evaluation or artifact verification completed at the same time.

9 Prospective 6-billion-token continuation

At the cutoff, a verified eight-RTX-5090 node was running, its GPU computation and environment checks had passed, and checkpoint/data preparation was in progress. **Real-data training had not yet been reported started.** Provider success or container-running status is not evidence of optimizer progress.

The approved design resumes the full 1.01 state at update 2250 and targets $T = 5720$. It requires additional pack-02 and pack-03 data, continuing disjoint document-hash intervals. R24 extends stream chaining and explicit checkpoint selection without an architectural change. The pack-02 boundary is zero-based update 3594; the pack-03 boundary must be computed from the actual built pack-02 capacity, not assumed from its nominal name. Before the long run, real-batch checks compare old and new stream selection and the early continuation against updates 2260, 2270, and 2280; the established loss-difference stop tolerance is 0.005 nats per token. A separate first-100-update throughput check requires at least 95,000 tokens/s.

The planned extension requires 3,638,558,720 new token presentations after its resume point. This exceeds the difference between the released versions' terminal token counters because the cooled 1.01 tail is not inherited.

The intended decay boundary is $q = 4576$; its first lower-rate completed update is 4578. Checkpoints are retained every 150 updates, additionally at 4576 and final update 5720: 25 new full states in total. At the observed native-state size, history alone would occupy 221,507,178,775 bytes. Training streams, the resume state, current and temporary checkpoints, caches, and export require additional space; 400 GB was allocated rather than relying on deletion of unverified states. No result, final checkpoint, benchmark gain, or validated loss forecast is claimed for this continuation.

10 Reproducibility and interpretation checklist

A reproduction should pin the dataset revision, tokenizer bytes, trainer, loader, model weights, optimizer state, global batch, stream boundaries, and completed-step numbering. It should verify actual downloaded artifacts, not only metadata, and preserve per-item outputs so that paired comparisons remain auditable. Native inference weights and resumable optimizer states are different artifact types. Benchmark implementations, context policies, normalization, language, and base/instruct status must accompany every score.

The present evidence is one training lineage with a small tokens-per-parameter budget. Its strongest conclusion concerns measured continuation behavior. General architectural rankings, human preference, robustness to paraphrase, and safe or reliable interaction require other evidence. Additional benchmark catalogues are useful sources of diagnostics, but a catalogue entry, pilot, full-split result, and eligible leaderboard submission are distinct states.

References

[1] SlayerLab. *Slayer1B 1.00: model card and release artifacts*. https://huggingface.co/SlayerLab/Slayer1B-1.00.

[2] SlayerLab. *Slayer1B 1.01: model card, inference package, and measured history*. https://huggingface.co/SlayerLab/Slayer1B-1.01.

[3] SlayerLab. *Slayer1B 1.00 resumable checkpoints*. https://huggingface.co/SlayerLab/Slayer1B-1.00-checkpoints.

[4] SlayerLab. *Slayer1B 1.01 resumable checkpoints and run records*. https://huggingface.co/SlayerLab/Slayer1B-1.01-checkpoints.

[5] SlayerLab. *Polish-dominant pack dataset*, revision d50df04a6ad46876fa559a87cca5d3e2a96e7098. https://huggingface.co/datasets/SlayerLab/slayer-pl-8x3b.

[6] Fabryka AI Track. *Versioned model results and run histories*. https://track.fabryka.ai/models. Primary run IDs: 48744ed2-3155-4294-b899-a6fda1f851da (1.00), 4934429a-d393-42e0-8aac-03eae6b71ff7 (1.01).

[7] Howe Tissue, Venus Wang, and Lu Wang. *Scaling Law with Learning Rate Annealing*. 2024. arXiv:2408.11029. https://arxiv.org/abs/2408.11029. Used for conditional training-loss scenarios, not benchmark prediction.

[8] CodeSOTA. *Under 1B: useful small models*. https://beta.codesota.com/sub-1b. Catalogue and versioned execution scopes, accessed 23 October 2026; the strict size class does not include this 1.074B model.